

WASP-140b

R-2189

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_211016 · created 2026-08-13T23:38:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459503.8537 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.044
Transit depth (Rp/Rs)^2	2.89 +/- 1.5 [%]
Orbital Inclination (inc)	82.61 +/- 1.54
Airmass coefficient 1 (a1)	0.9884 +/- 0.0086
Airmass coefficient 2 (a2)	0.0067 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	#3 - [414, 83]
Optimal Method	PSF photometry
Transit Duration (day)	0.043 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.17 ± 0.044 vs 0.1432 (deviation 0.42σ)
Duration fit vs published	0.043 d vs 0.0604 d (deviation 0.37σ)
Mid-time O–C	-6.0 min
Depth SNR	2.7
Residual scatter	0.87 %

Light curve

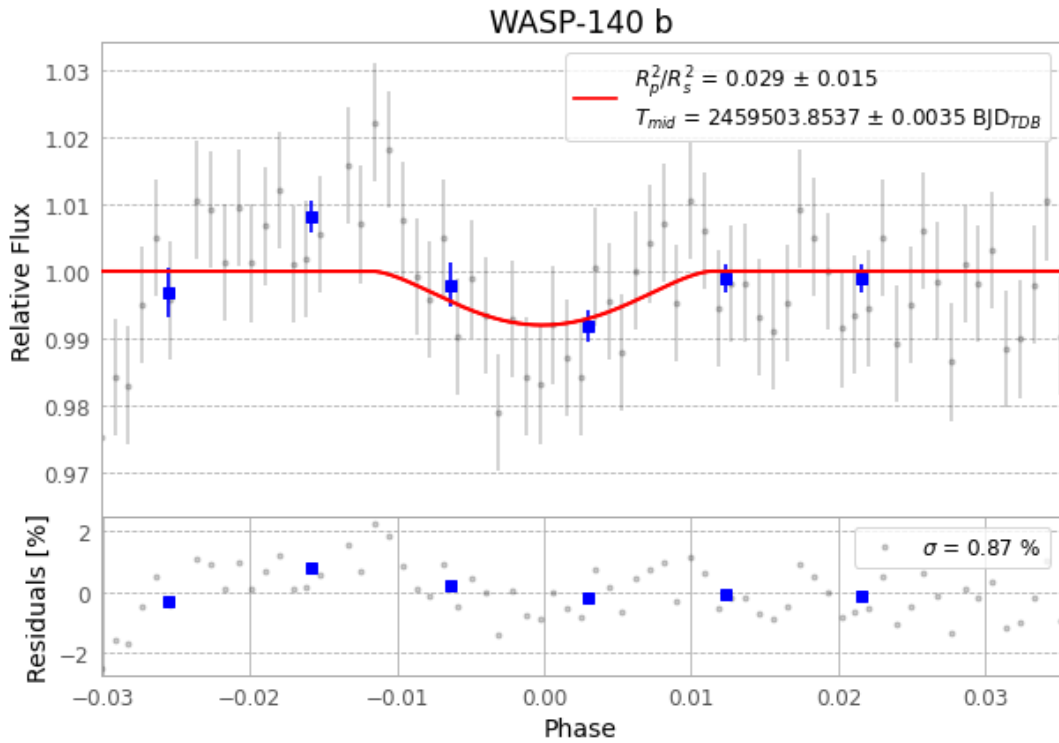


Figure 1 — FinalLightCurve_WASP-140 b_2021-10-15.png (SHA-256 ccf3b75bb3afa265...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [433, 51], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d001c212b93f747...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

71 FITS frames (47 MB), WASP-140211016064514.FITS → WASP-140211016101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-211016.log (606c138c4176...), FinalLightCurve_WASP-140 b_2021-10-15.png (ccf3b75bb3af...), FinalLightCurve_WASP-140 b_2021-10-15.csv (9ee4e7f680b6...)

Compute resources

Wall time 133.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:38Z.